

Learning activity 5: Ratio Estimators with Real Census Data

Using IPUMS ACS 2024 Data for New York State

Introduction

In class, we studied the **ratio estimator** under simple random sampling (SRS). In this activity, you will apply that theory to real survey data from the American Community Survey (ACS), accessed through [IPUMS USA](#).

The key idea behind the ratio estimator is that if we know the population total of some auxiliary variable x , and x is correlated with our variable of interest y , we can use that relationship to get a better estimate of the population total of y .

The ratio estimator for the population total is:

$$\hat{y}_r = \hat{B} \cdot t_x \quad \text{where} \quad \hat{B} = \frac{\bar{y}_S}{\bar{x}_S}$$

and t_x is the **known** population total of x . In this activity we use **annual hours worked** as our auxiliary variable and **wage income** as our variable of interest. We treat the full filtered ACS dataset as the population, draw an SRS from it, and then verify our estimates against the true population values.

Setup

```
library(tidyverse)
library(ipumsr)
library(survey)
```

```
ddi <- read_ipums_ddi("usa_00011.xml")
data <- read_ipums_micro(ddi)
```

Prepare the Population

We filter the ACS data to private wage and salary workers in New York State with positive hours and wage income, removing the top and bottom 5% of earners to reduce the influence of extreme outliers. We then construct **annual hours worked** as our auxiliary variable. For this question we will consider this the population.

```
ny_pop <- data |>
  filter(
    STATEFIP == 36,      # New York
    UHRSWORK > 0,
    INCWAGE > 0,
    INCWAGE < 999998,    # 999998/999999 are IPUMS N/A codes
    WKSWORK1 > 0,
    CLASSWKR == 2        # private wage/salary workers
  ) |>
  filter(
    INCWAGE < quantile(INCWAGE, 0.95),
    INCWAGE > quantile(INCWAGE, 0.05)
  ) |>
  mutate(annual_hours = UHRSWORK * WKSWORK1)

nrow(ny_pop)
```

```
[1] 86436
```

Draw an SRS and Check the Key Assumption

We drew an SRS of size 1000 from the population. The ratio estimator requires a **proportional relationship** between y and x .

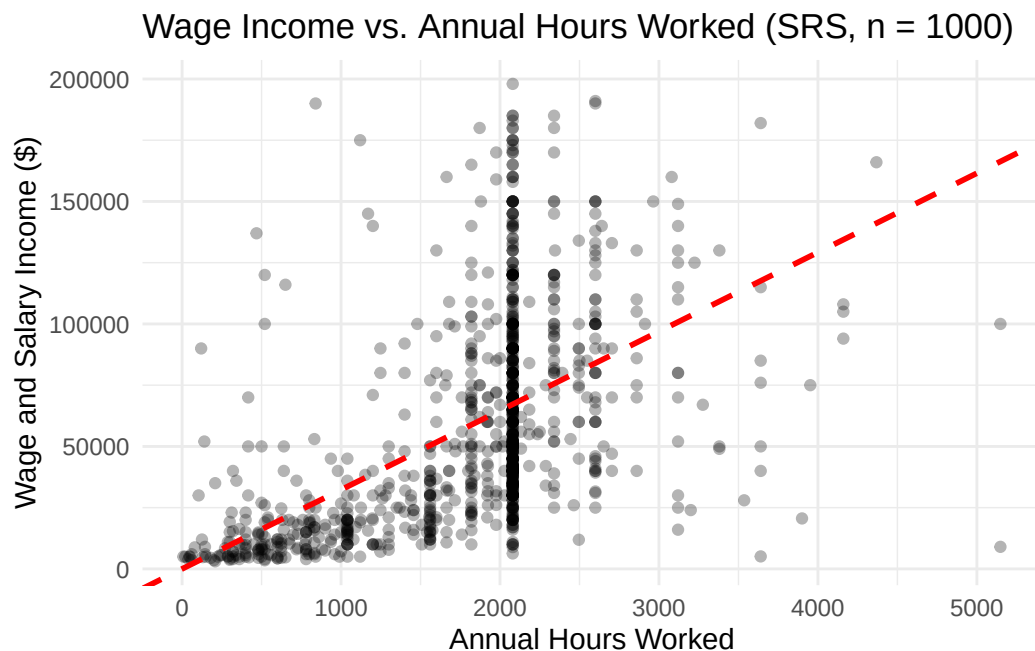
```

set.seed(123)
ny_sample <- ny_pop |> sample_n(1000)

B_hat = mean(ny_sample$INCWAGE)/mean(ny_sample$annual_hours)

# Scatterplot
ggplot(ny_sample, aes(x = annual_hours, y = INCWAGE)) +
  geom_point(alpha = 0.3) +
  geom_abline(intercept = 0, slope = B_hat, col = "red", lwd = 1, linetype="dashed")+
  labs(
    title = "Wage Income vs. Annual Hours Worked (SRS, n = 1000)",
    x     = "Annual Hours Worked",
    y     = "Wage and Salary Income ($)"
  ) +
  theme_minimal()

```



Question 1: Examine the scatterplot. Comment on whether or not you think the proportionality assumption is reasonable here. Comment on the strength of the correlation between hours worked and income.

- The relationship between hours worked and wage is reasonably proportional here. It is a relationship through the origin, in the sense that if a person worked 0 hours they will make 0 wage. The relationship is not perfectly linear especially for annual hours below 2000 hours.

- The correlation between hours worked and wage is positive but is also weak. We observe this because there is a large dispersion of points around the fitted line (red).
- The correlation is stronger for shorter amount of hours work and weaker for longer amount of hours worked.

Efficiency of ratio estimation

The ratio estimator is more efficient than the sample mean estimator under SRS when:

$$R > \frac{1}{2} \cdot \frac{CV(x)}{CV(y)}$$

where CV is the coefficient of variation and R is the population correlation. Check whether this condition holds:

```
cv_x      <- sd(ny_pop$annual_hours) / mean(ny_pop$annual_hours)
cv_y      <- sd(ny_pop$INCWAGE)       / mean(ny_pop$INCWAGE)
threshold <- 0.5 * (cv_x / cv_y)
R         <- cor(ny_pop$annual_hours, ny_pop$INCWAGE)
```

Question 2: Does the efficiency condition hold? Interpret what this means in plain language.

- Here $R = 0.51$ and the threshold is 0.27. Therefore under this threshold condition, the ratio estimator is more efficient than the sample mean estimator under SRS. That is, it will have a smaller MSE than the sample mean estimator under SRS.
- The linear correlation coefficient is fairly strong 0.51.
- The coefficient of variation for x is 0.39 and for y is 0.71. This means that income varies much more, relative to its mean than hours worked. This indicates that hours worked will explain some of the variation in income but likely not a lot of it. Ideally, we would like the ratio of coefficient of variations to be close to 1. In this dataset, the ratio is 0.524 indicating that there is unexplained variations in income that is not explained by hours worked.

Estimate the Ratio and Construct a Confidence Interval

We estimate $\hat{R}$ and construct a 95% CI under SRS using standard error formula:

$$\hat{V}(\hat{B}) = \left(1 - \frac{n}{N}\right) \frac{s_e^2}{n\bar{x}_S^2}$$

where $s_e^2 = \frac{1}{n-1} \sum_{i \in S} (y_i - \hat{B}x_i)^2$

```
n      <- nrow(ny_sample)
N      <- nrow(ny_pop)
fpc    <- 1 - n / N
B_hat  = mean(ny_sample$INCWAGE)/mean(ny_sample$annual_hours)
xbar   = mean(ny_sample$annual_hours)

se2    = 1/(n-1)*sum((ny_sample$INCWAGE - B_hat*ny_sample$annual_hours)^2)

Vhat   = se2/(n*xbar^2)*fpc
SE     = sqrt(Vhat)

ci_upr = B_hat + qt(0.975, df = 1000-1)*SE
ci_lwr = B_hat - qt(0.975, df = 1000-1)*SE
```

Question 3: Interpret and give an estimate for B in plain language and give a 95% confidence interval for B .

- The estimate for the income per hours worked is 32.3.
- A 95% confidence interval for the income per hours worked is [31.0, 33.6]. (I rounded up)

Verification of results using the survey package

```
# Set up SRS design
svy <- svydesign(
  ids      = ~1,          # no clustering
  fpc      = ~N,          # finite population correction
  data     = ny_sample |> mutate(N = nrow(ny_pop))
)

# Ratio estimate
```

```
ratio_svy <- svyratio(
  numerator   = ~INCWAGE,
  denominator = ~annual_hours,
  design      = svy
)

print(ratio_svy)
```

```
Ratio estimator: svyratio.survey.design2(numerator = ~INCWAGE, denominator = ~annual_hours,
  design = svy)
```

```
Ratios=
```

```
      annual_hours
INCWAGE      32.30575
```

```
SEs=
```

```
      annual_hours
INCWAGE      0.6456256
```

```
confint(ratio_svy, df = 999)
```

```
              2.5 %   97.5 %
INCWAGE/annual_hours 31.03881 33.57269
```

Question 4: This is code to complete Q3 using the survey package. Verify that the estimate and the confidence interval you computed in Question 3 matches the one using this R package. (It should!)

Ratio Estimator for the Population Total

Now we use $\hat{R}$ to estimate the **total wage income** of all private wage/salary workers in New York, and construct a CI for the total.

```
t_x <- sum(ny_pop$annual_hours) # known population total of annual hours
t_y <- sum(ny_pop$INCWAGE)      # true total income (for verification)

# Point estimate
t_hat_r <- B_hat * t_x

# Variance of total:

n      <- nrow(ny_sample)
```

```

N      <- nrow(ny_pop)
fpc    <- 1 - n / N
B_hat = mean(ny_sample$INCWAGE)/mean(ny_sample$annual_hours)
xbar = mean(ny_sample$annual_hours)

se2 = 1/(n-1)*sum((ny_sample$INCWAGE - B_hat*ny_sample$annual_hours)^2)

Vhat = se2/n*(t_x/xbar)^2*fpc
SE = sqrt(Vhat)

t_hat_r + qt(0.975, df = 1000-1)*SE

```

```
[1] 5267856529
```

```
t_hat_r - qt(0.975, df = 1000-1)*SE
```

```
[1] 4870268621
```

Question 5: What is the estimate for total wage income and a 95% CI? Does your 95% CI contain the true population total?

- The estimate for the the total annual wage is \$5,069,062,575 or \$5.07 billion.
- A 95% confidence interval for the total annual wage is [\$4.87, \$5.27] billion

Verification of results using the survey package

```

# Set up SRS design
svy <- svydesign(
  ids      = ~1,          # no clustering
  fpc      = ~N,          # finite population correction
  data     = ny_sample |> mutate(N = nrow(ny_pop))
)

# Ratio estimate
ratio_svy <- svyratio(
  numerator   = ~INCWAGE,
  denominator = ~annual_hours,
  design      = svy
)

```

```
predict(ratio_svy, total = t_x)
```

```
$total
      annual_hours
INCWAGE 5069062575

$se
      annual_hours
INCWAGE 101304466
```

Question 6: This is code to complete Q6 using the survey package. Verify that the estimate and the standard error you computed in Question 6 matches the one using this R package. (It should!)

Question 7: In this activity we used the full ACS dataset as the “known” population total t_x . In a real application where you only have a sample, where could you get t_x ?

- You could get this information from a country’s census (statistics canada for Canada)
- Past reliable data. If you have a previous wave of the same survey or a larger companion survey, its estimated total can be used. Ideally, you would combine its uncertainty as well into your model.
- You could use labor force statistics data (statistics canada for Canada)
- The Census Bureau releases official totals for things like total employment, hours worked, and earnings at the national, state, and county level through the ACS summary tables, decennial census, or the Current Population Survey.
- In countries with population registers (Scandinavia etc.), exact totals for many variables are simply known from government databases.
- IRS have aggregate totals derived from payroll records, tax filings, or employer reports that cover the full population.

Submission

Please submit this .qmd file rendered as a PDF, with all code chunks run and all questions answered. Make sure your file renders without errors before submitting.